

Linear Algebra, Lecture 3: Linear maps, matrices and determinants

3.1 Basis vectors and linear combinations

Every vector v in $\mathbb{R}^2$ can be written uniquely as $v = x e_1 + y e_2$, where $e_1 = (1, 0)$ and $e_2 = (0, 1)$ are the standard basis vectors (also written $\hat{i}$ and $\hat{j}$). The numbers x and y are the coordinates of v . A sum of scaled vectors $a u + b w$ is called a linear combination of u and w . The set of all linear combinations of u and w is their span; if u and w are not parallel, the span is the whole plane.

Example: $(3, 2) = 3 e_1 + 2 e_2$.

3.2 Linear maps

A map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear if $T(u + w) = T(u) + T(w)$ and $T(c u) = c T(u)$ for all vectors u, w and scalars c . Geometrically, a linear map keeps grid lines parallel and evenly spaced and keeps the origin fixed.

Because $v = x e_1 + y e_2$, linearity gives $T(v) = x T(e_1) + y T(e_2)$. So T is completely determined by the two vectors $T(e_1)$ and $T(e_2)$. Writing them as the columns of a 2×2 matrix $A = [T(e_1) \ T(e_2)]$ we get

$$A v = x (\text{first column of } A) + y (\text{second column of } A).$$

Example: rotation by 90 degrees sends e_1 to $(0, 1)$ and e_2 to $(-1, 0)$, so its matrix is $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Then $R(3, 2) = 3(0, 1) + 2(-1, 0) = (-2, 3)$.

3.3 Matrix multiplication is composition

If we first apply A and then B , the result $v \rightarrow B(A v)$ is again linear. Its matrix is the product BA , defined so that $(BA) v = B(A v)$ for every v . Reading off columns: the first column of BA is B times the first column of A , and the second column of BA is B times the second column of A . Note the order: BA means "first A , then B ".

Example: with the rotation R above and the shear $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $SR = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$ while $RS = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$. In general $AB \neq BA$: matrix multiplication is not commutative, because doing two motions in a different order gives a different result.

3.4 The determinant

A linear map scales all areas by the same factor. The unit square spanned by e_1 and e_2 is mapped to the parallelogram spanned by the columns of A ; its signed area is the determinant $\det A = ad - bc$ for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $\det A$ is negative the map flips orientation; if $\det A = 0$ the plane is squashed onto a line or a point and the map cannot be undone.

Since areas are scaled first by $\det A$ and then by $\det B$, we get $\det(BA) = \det B \det A$.

Examples: $\det R = 1$ (rotations keep areas), $\det S = 1$ (shears keep areas), $\det \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} = 6$.

Exercises

1. Compute RS and SR for the matrices above and check that they differ.
2. Find a 2×2 matrix with determinant 0 and describe what it does to the plane.